CloudTrail, CloudWatch, and Config Lab

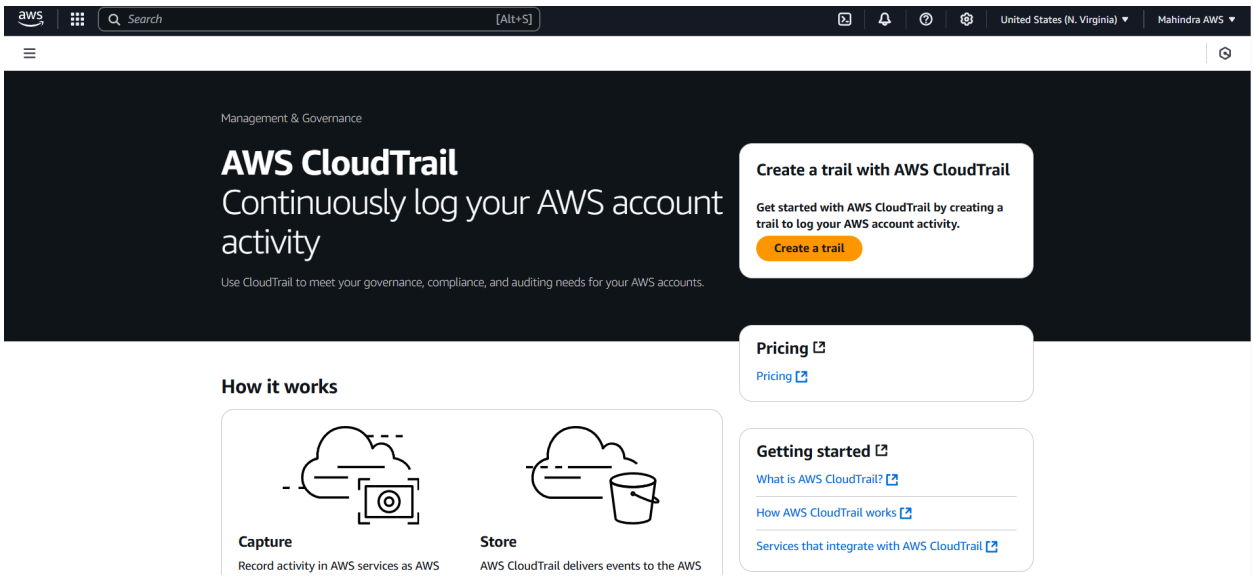
Author : Mahindra Guptha Kotha
Mail ID: [redacted]

EXERCISE 7.1

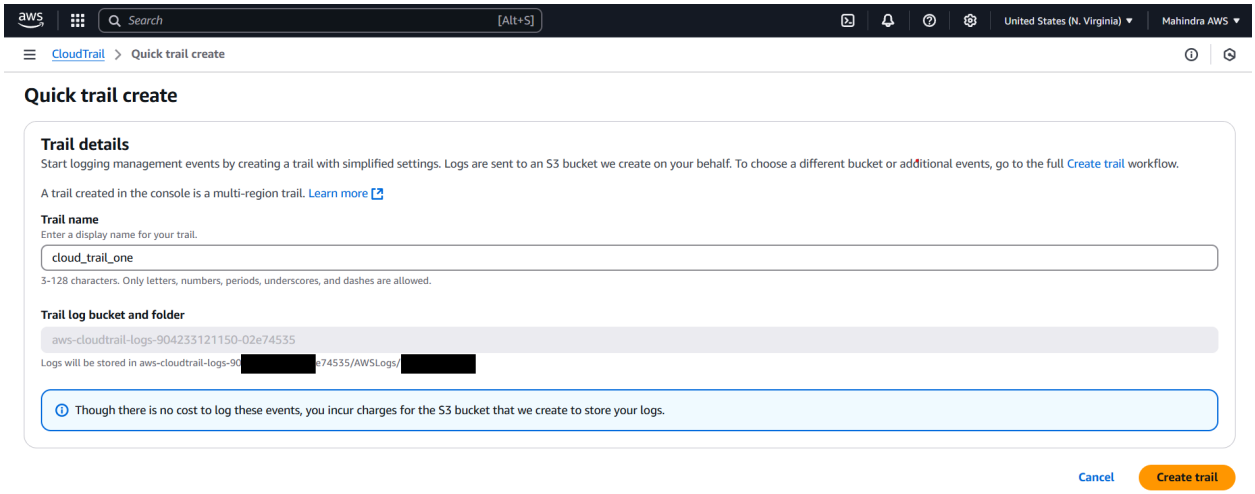
Create a Trail

In this exercise, you'll configure CloudTrail to log write-only management events in all regions.

1. Browse to the CloudTrail service console and click the Create Trail button.

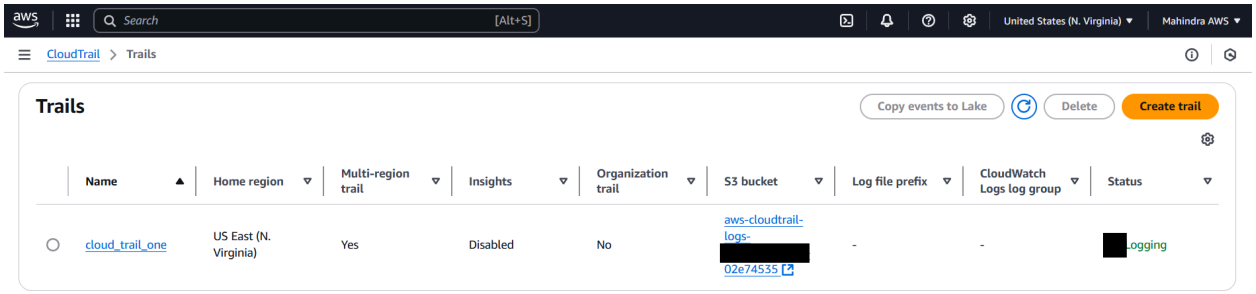


2. Under Trail Name, enter a trail name of your choice. Names must be at least three characters and can't contain spaces.



3. Under the Storage Location heading, select Create New S3 Bucket. Enter the name of the S3 bucket you want to use. Remember that bucket names must be globally unique.
4. Under Log File SSE-KMS Encryption, clear the box next to Enabled.
5. Enter a custom name for the AWS KMS Alias.
6. Leave all other settings at their defaults and click the Next button.
7. Under Event Types, select the box next to Management Events. Don't select any other boxes.
8. Under Management Events, make sure only Write is selected.
9. Click Next.
10. Review the settings and click the Create Trail button.

All the above steps are handled by AWS recent update so I need not to select S3 bucket and details it will create a S3 bucket under log file.

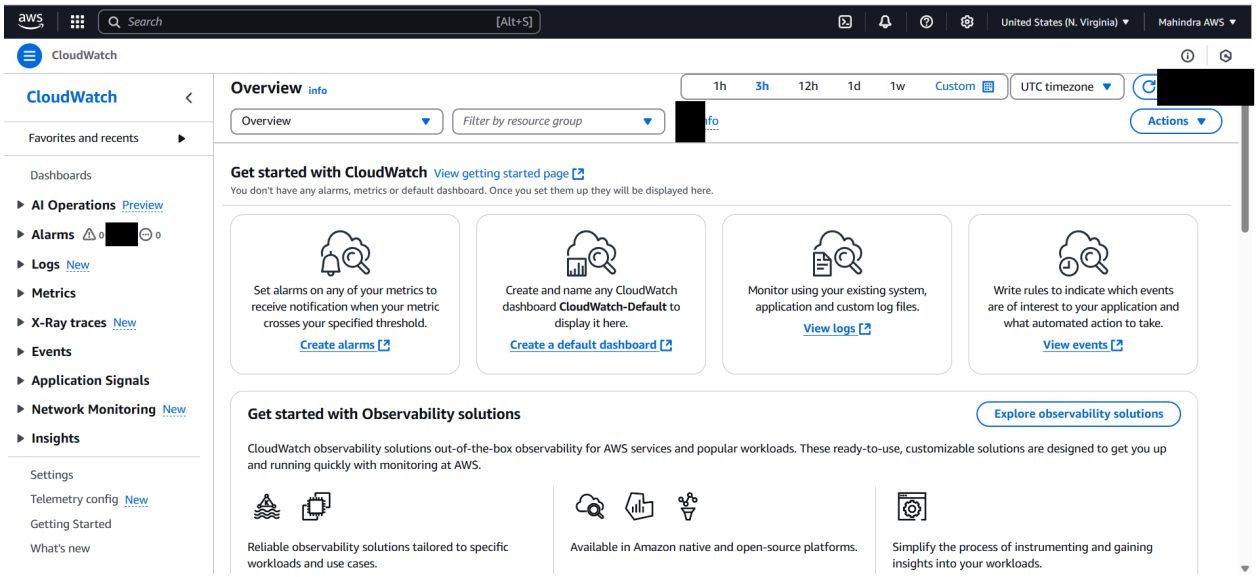


EXERCISE 7.2

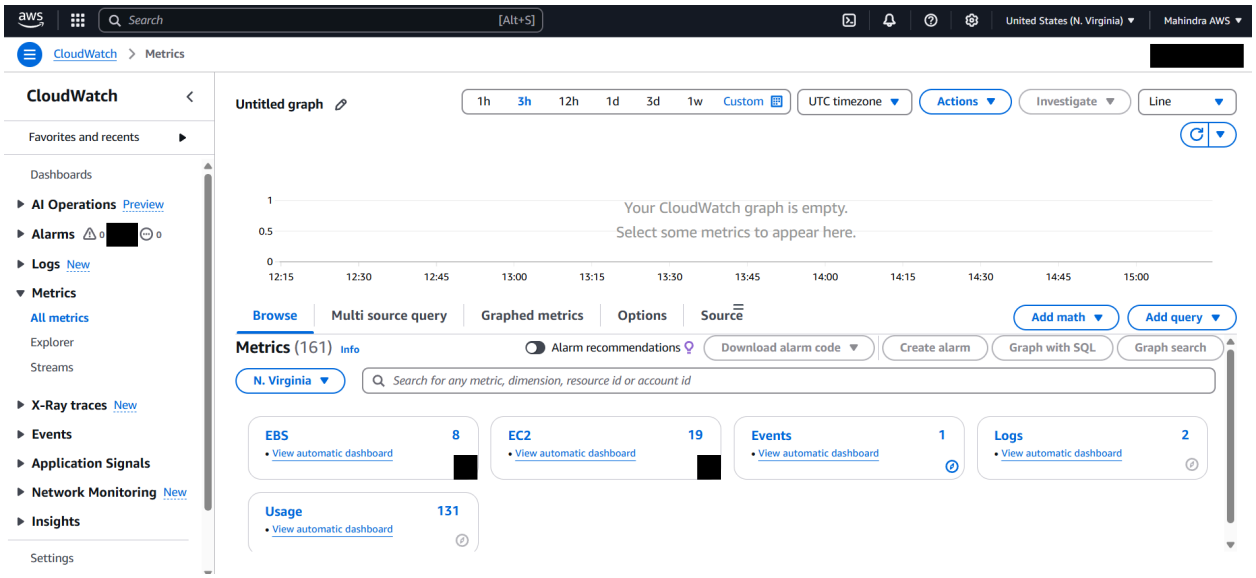
Create a Graph Using Metric Math

In this exercise, you'll create a graph that plots the `NetworkIn` and `NetworkOut` metrics for an EC2 instance. You'll then use metric math to graph a new time series combining both metrics.

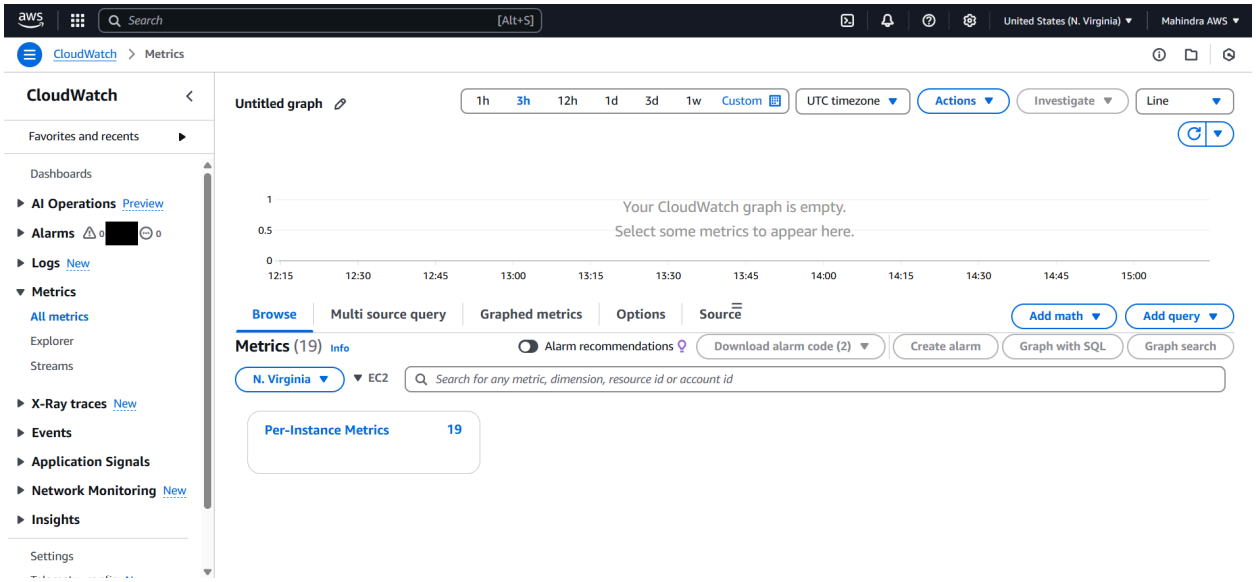
1. Browse to the CloudWatch service console and expand Metrics on the navigation menu.

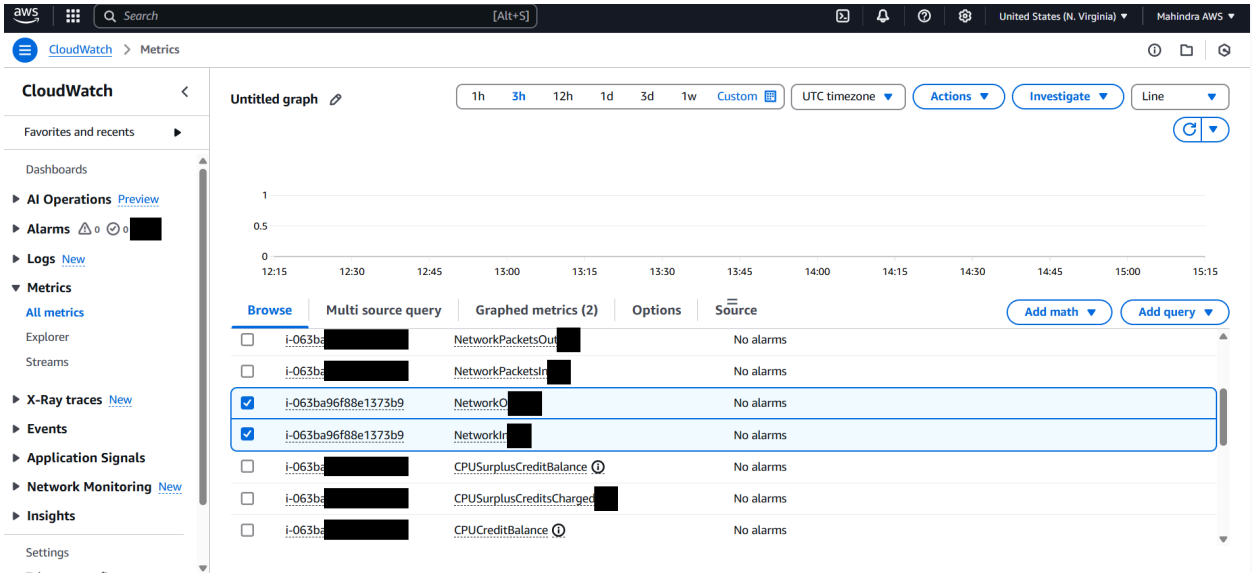


2. Click All Metrics.

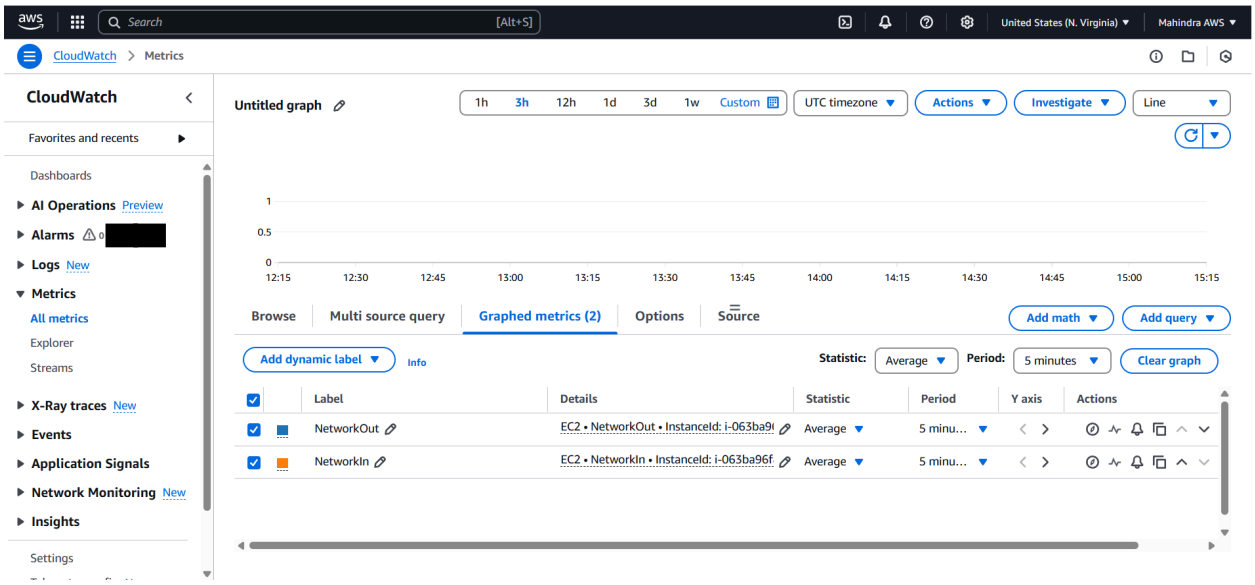


3. On the Browse tab, descend into the `EC2` namespace. Select Per-Instance Metrics; then locate and select the `NetworkIn` and `NetworkOut` metrics.

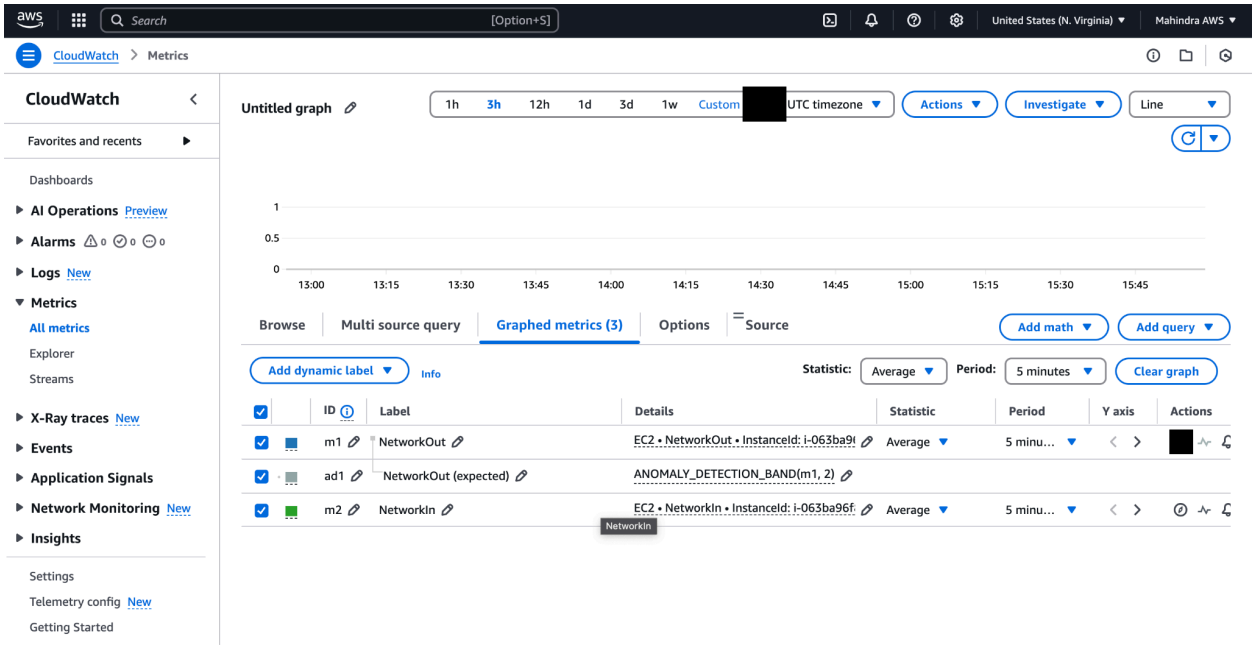




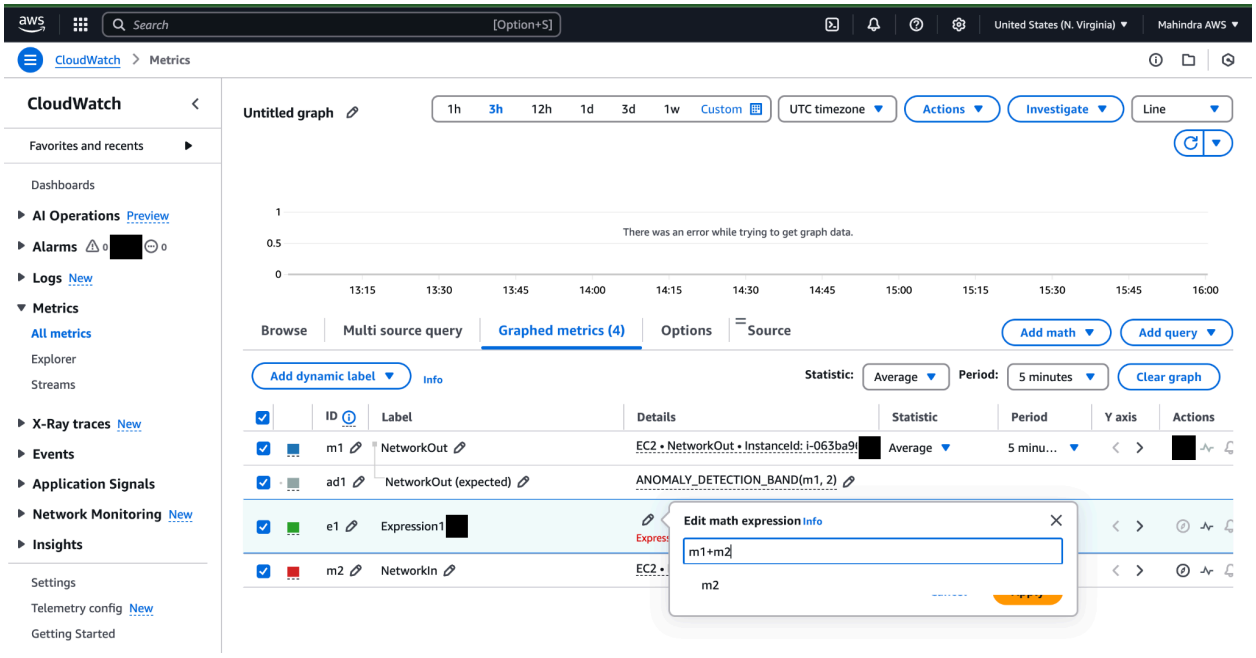
4. Click the Graphed Metrics tab.



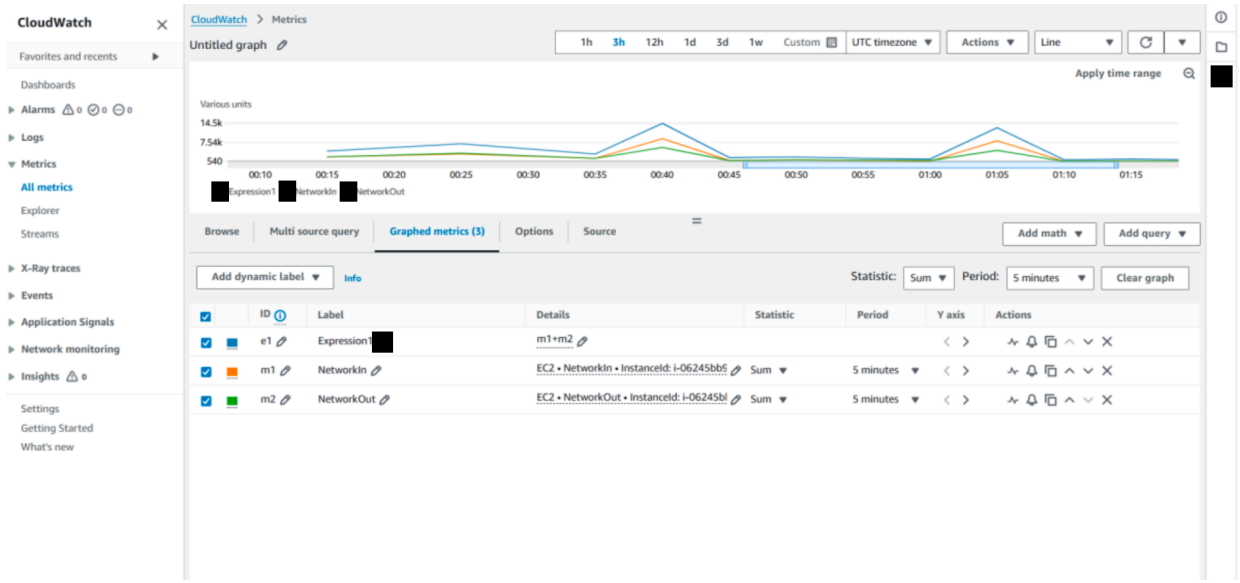
5. For each metric, select Sum for Statistics and 5 Minutes for Period. Refer to Figure 7.2 as needed.



6. Click the Add Math button and select Start With Empty Expression.
7. In the Edit Math Expression field, enter the expression **m1+m2**.



8. Click the Apply button. CloudWatch will add another time series to the graph representing the sum of the `NetworkIn` and `NetworkOut` metrics. Your graph should look similar to this one:

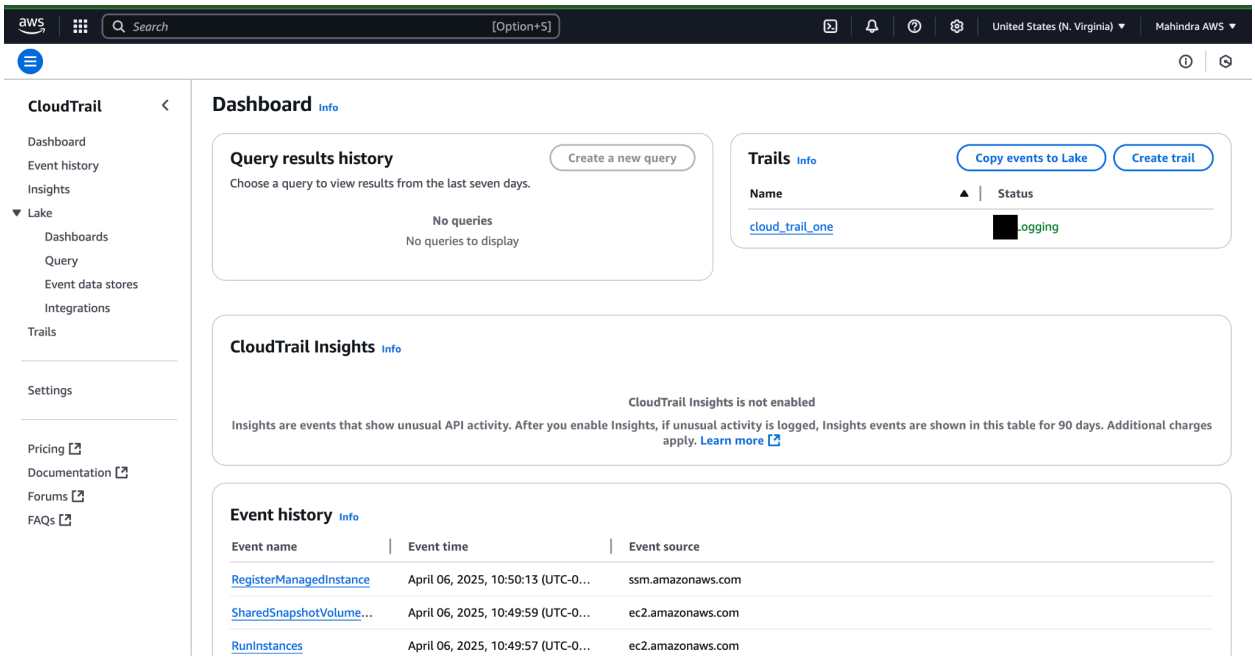


EXERCISE 7.3

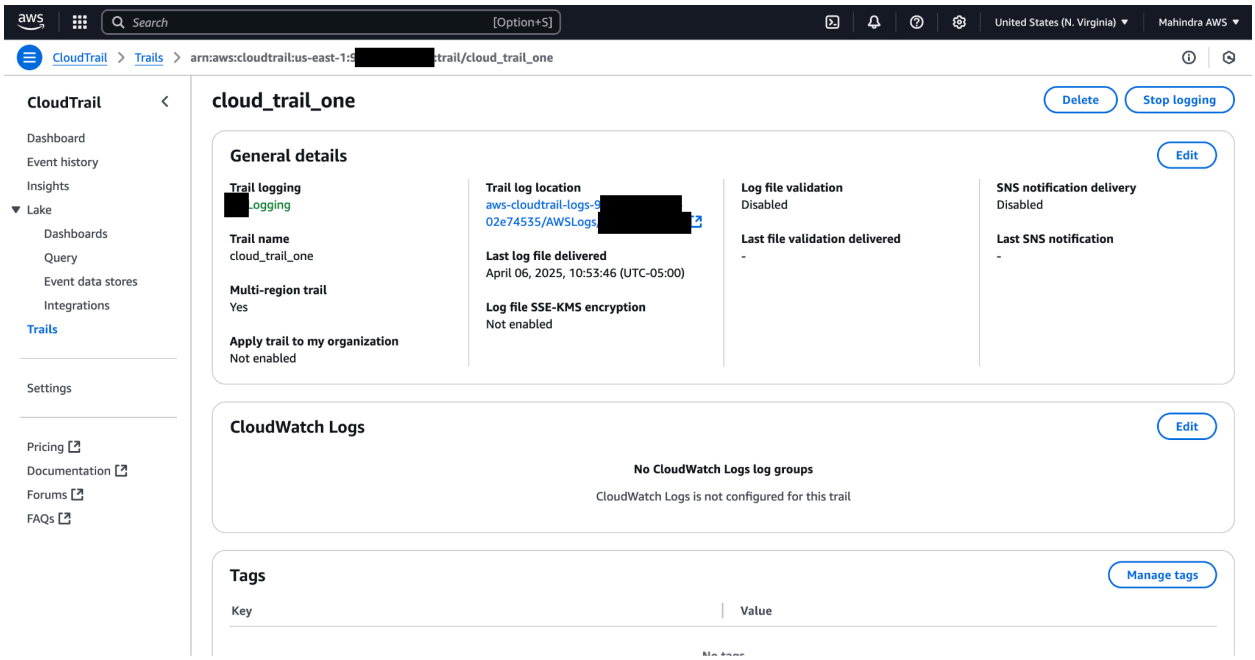
Deliver CloudTrail Logs to CloudWatch Logs

In this exercise, you'll reconfigure the trail you created in Exercise 7.1 to stream events captured by CloudTrail to CloudWatch Logs.

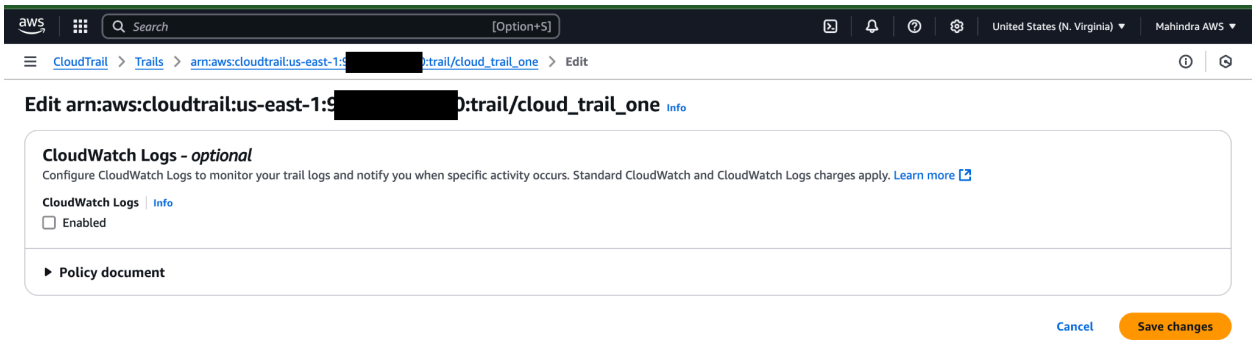
1. Browse to the CloudTrail service console and click Trails.



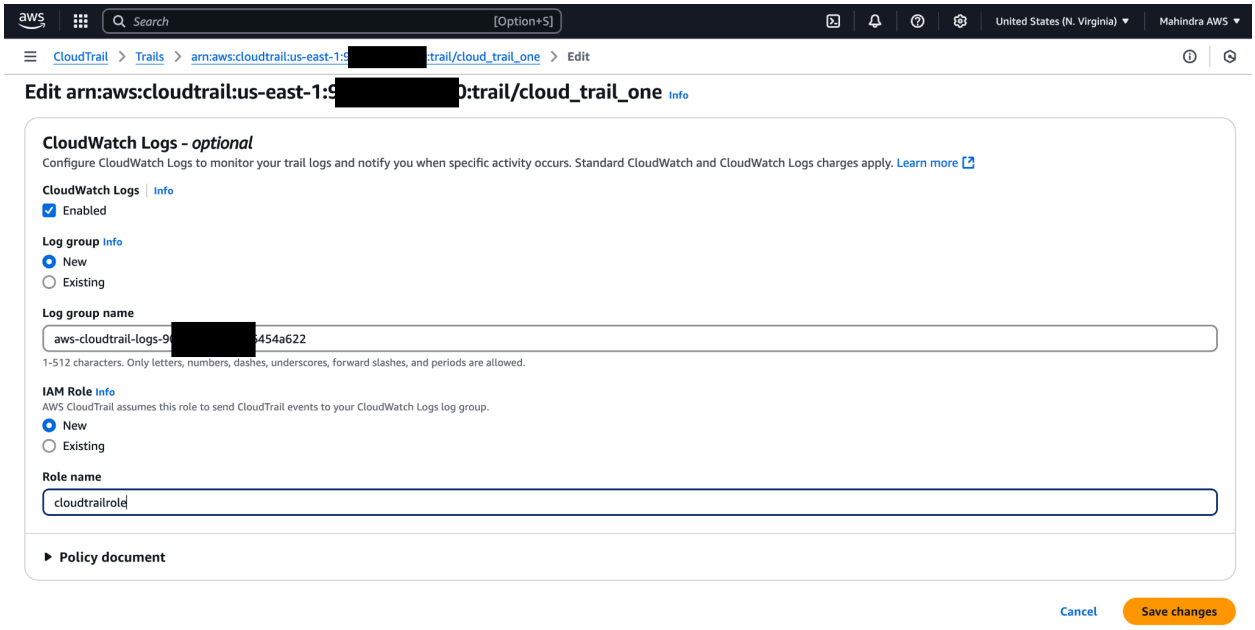
2. Click the name of the trail you created in Exercise 7.1.



3. Under the heading CloudWatch Logs, click the Edit button.



4. Under CloudWatch Logs, select the Enabled check box.
5. CloudTrail prompts you to use a New or Existing Log Group. Select New and enter a log group name of your choice.
6. CloudTrail must assume an IAM role that will give it permissions to stream logs to CloudWatch Logs. CloudTrail can create the role for you. Just click the New radio button under IAM Role. Enter a custom role name of your choice.



7. Click the Save Changes button.

